niques which do this — but the effect of bumps and indentations are created when the object is rendered. Bumps and depressions reflect light differently, and this is used to give the required effect. The rock in the lower right of the image of a lady standing on rocky ground in front of trees is a good example here. There is a texture map to give the colours and patterns in the rock, and a bump map to give the rock a rough-looking surface. In that image, the area to the left has not been rendered, and you can see the basic triangles from which the scene is constructed. Without bump mapping, a very much larger number of triangles would be needed to give a similar effect.

Another way of creating these effects is procedural mapping. Here, a mathematical function, or procedure, is used to create the required effect, instead of a raster image. This would not be appropriate for the logo on the side of an aeroplane — there is really only one way to do that — but it would be very suitable, for, say, the patterns in granite, or the ripples on the surface of water. One advantage of procedural mapping is its programmability. Just by changing the basic parameters — numbers — the resulting map changes; e.g. the granularity of the granite, the colour of and contrast between the grains. Also, these parameters can be changed throughout an animation to give special effects. So, a procedural map can be fine tuned in ways that are impossible with normal image-based maps. Mind you, instead of a single image, whole animations or videos can be used as a texture map, giving endless possibilities.

The two other vital components in any scene are cameras and lights. The camera is a fairly obvious concept, and is the viewpoint from which the scene is rendered. Naturally, you have considerable control over various parameters associated with any cameras — the effective focal length, position, angle of tilt, and so forth. All of these can also be animated.

The lighting in a scene is one of the most difficult things to get right, and deserves close study for anybody serious about modelling and animation. Basic lighting consists of ambient light — a general light present, such as the light from a cloudy sky — and then specific lights placed in a scene. Again, there is some considerable variety in the control you have over these lights, but this is not quite enough for realistic effects. A scene using standard lighting techniques simply looks rendered. Those images that are the most realistic use advanced techniques such as ray tracing and radiosity.

Both of these methods employ the physics of light. In ray tracing, each ray of light is followed back from the viewpoint to its origin, in order to determine its colour and intensity. For example, looking at a painting on a wall, following back to a point on that painting, there will be a particular colour in the painting itself; the light falling on it may be composed of ambient light streaming in through the windows covered with yellow curtains, and this light will be reflecting off the pale blue walls of the room. The point here is that the colour and intensity of any ray of light depends on many factors, simply due to the fact that light is reflected from many surfaces, passes through some transparent materials, each of these subtly affecting the final colour. Ray tracing takes as many of these factors into consideration as possible. This method is also particular good when reflective surfaces such as mirrors are involved.

Radiosity also uses basic physics to combine together the effects of multiple light sources and surfaces, but it does so in a general sense, and not from the point of view of a camera. The results are added to the scene itself, rather like a general bitmap that is applied all over. Effects such as shadows, light coming through yellow curtains, and so forth, are handled well by this method, but it is only really suitable for diffuse surfaces, and not reflective surfaces — if the camera moves, the reflection will change, and so you will need to use a reflection map in such instances.

The big problem with radiosity is that if an object in the scene moves, then the results of the radiosity calculations are broken. These can take quite some time to compute — it is an iterative process, and the quality improves the longer the time you give the system. So, this method is suitable for applications such as an architectural fly-through of a building or other structure. The radiosity calculations are done just once, and then the camera is animated and moved through the scene. The calculations needed for radiosity are generally more compute intensive than ray tracing, and so when objects in the scene are animated, ray tracing is the preferred method.

The description that we have given here is from the point of view of software such as 3ds MAX. But there are other ways of representing objects in a scene. Maya is particularly good with what is called NURBS content. NURBS stands for non-uniform rational B-splines, and this is a method whereby curves and surfaces are represented mathematically rather than by collections of triangles; you can think of the comparison here between image-based maps and procedural maps. This is a procedural way of representing surfaces.

Digital Passion | **Lead Feature**

Usually, the NURBS method uses much less data than triangles. If you're building a 3D sphere with triangles you will need a high number of faces, otherwise it will look too jaggy. But if you define the sphere as a single surface based on a mathematical formula it will be much simpler in terms of raw data.

This method is particularly suitable for organic surfaces — modelling a human figure from triangles is a difficult procedure, and NURBS gives you a different level of control that is much more suitable for such objects. It is also useful for smooth and perhaps complex surfaces such as the aeroplane on which the logo has to go.

The specialities of these different programs make them more or less attractive in different fields.

Software like 3ds MAX is better for architectural or low-level polygon scenes, and it is this that is important for anyone developing applications that render in real-time, such as video games, virtual worlds, etc. Two points here: the most successful CAD program on the PC is AutoCAD, also by Autodesk, and as MAX and its predecessor 3D Studio have long been compatible with AutoCAD file formats, MAX has naturally been popular with AutoCAD users, such as architects and engineers. Also, with real-time rendering, such as in a game, to enhance performance it is important to have as small a number of triangles in the scene as possible. Mind you, the latest graphics cards for gamers, with several hundred processor cores, are capable of rendering triangles at rates that would probably not have been believed, even just ten years ago.

MAX is also well set up for character rigging, layered animation, etc. In short, if you're doing gaming, visualisation, virtual worlds — anything real-time — then you're probably better off using something like MAX. Also, pre-rendered animations like architectural fly-throughs where radiosity can be used.

NURBS-based software such as Maya is probably better suited to high-detail material that will be pre-rendered. For example, if you're creating a static scene that has lots of complex surfaces, liquids, mist, fog, funky lighting, and does not need to be rendered or manipulated in real-time. Similarly, if you are creating high detail lush scenes with crashing waves, complex curvy shapes, weird lighting, particle effects, dinosaurs, and so on, then software like Maya is probably more appropriate. CGI (computer-generated imagery) movies are, of course, pre-rendered so you would expect Bollywood to use Maya more than the games industry, which would favour MAX.

Animation In India

The quality of the finished product has improved so dramatically during the last few years that you have probably seen some CGI in a movie and did not realise that it was computer generated. As a result, the demand for 3D modelling and animation around the world has boomed. It is not just for the likes of dinosaurs in *Jurassic Park*, and new applications are cropping up all the time: games and other computer simulations, titles and intros for TV programs and films, advertisements, educational animations — even some airline safety videos are animated rather than using real actors. Particularly with Bollywood, there is a significant local demand for modelling and animation, but with this worldwide boom, thanks to the lower cost structures in India, a large amount of work is flowing into this country.

The total animation industry here is thought this year to be worth close to Rs 1,600 crore, and to be growing at 15 to 20 per cent per year — some parts of the market are thought to be growing at around 30 per cent. The problem is that we have a major skills shortage. One estimate suggests that this year the demand for animation professionals in India is around 300,000 — a huge increase over previous years, and the various training centres can hardly keep up.

So, there are great career opportunities here, and clearly the large number of training courses available are trying to meet the huge demand. However, although attending a course is important, it is not enough, and you simply need to get a great deal of practice in both modelling and animation. If you do end up working for an animation company, you will most likely need to use 3ds MAX or Maya. So, to get some practice, you should nip down to the local market and pick up a copy. Well, maybe not — this software is very expensive, but there are alternatives available at no cost at all, and these can be quite suitable for developing experience and skills in modelling. Going for a job interview with a satisfactory pass in some course is all very well, but how much better to present an impressive animation or model that you have developed yourself in your spare time.

Blender

The best place to start might very well be the open source modelling and animation package Blender (**www.blender.org**). This is available in both 32- and 64-bit Windows and Linux versions and also for Mac OS X and Solaris. This has rich functionality, on a par with software such as MAX and Maya, although it also has a rather funky user interface — but that is